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Bioaccumulation and Detoxication of Nodularin in Tissues of Flounder (*Platichthys flesus*), Mussels (*Mytilus edulis, Dreissena polymorpha*), and Clams (*Macoma balthica*) from the Northern Baltic Sea

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Cyanobacterial hepatotoxin accumulation in mussels (*Mytilus edulis, Dreissena polymorpha*), clam (*Macoma balthica*), and flounder (*Platichthys flesus*) tissues was measured. Flounder were caught with gillnets from the western Gulf of Finland on 21 August 1999, 25 July 2000, and 25 August 2000. Blue mussels were collected from: (1) a steel cage at a depth of 3 m on 20 August 1999, (2) an enclosure at depths of 3–5 m, and (3) an artificial reef (wreck at 25–30 m) in the western Gulf of Finland between June and September 2000. Furthermore, blue mussels were collected from two sites between August and October 2000: south of the town of Hanko at depths of 5 and 20 m in the western Gulf of Finland and south of the city of Helsinki at a depth of 7 m in the central Gulf of Finland. *M. balthica* and *D. polymorpha* were collected at a depth of 12 m from Russian waters in the eastern Gulf of Finland on 1–4 August 2000. The samples were analyzed for the cyanobacterial hepatotoxins nodularin (NODLN) and microcystins (MCs) using enzyme-linked immunosorbent assay (ELISA), liquid chromatography–mass spectrometry (LC-MS), and matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS). ELISA indicated a time-dependent accumulation of hepatotoxins in flounder liver up to 400 ± 10 (SD) μg/kg on 25 August 2000. No hepatotoxins were detected in flounder muscle samples. In blue mussels, collected from an enclosure 3–5 m deep in the western Gulf of Finland on 23 August 2000, ELISA indicated cyanobacterial hepatotoxins up to 1490 ± 60 μg/kg dry wt. Blue mussels collected from the other sites contained less cyanobacterial hepatotoxins (40–130 μg/kg dry wt). Clams and mussels from Russian waters contained cyanobacterial hepatotoxin at about 100–130 μg/kg dry wt. Total hepatotoxin levels in mussels from enclosures decreased from August to September, indicating at least partial detoxication/depuration of the toxins. LC-MS verified the presence of NODLN in mussels and flounder. Typical detoxication conjugates were observed by MALDI-TOF-MS in mussel samples collected during August 2000. In deeper-living wreck mussels cyanobacterial hepatotoxin levels continued to increase, from August to September, indicating that portions of cyanobacterial hepatotoxins reach the sea floor. NODLN bioaccumulation is a constant phenomenon in the area. © 2002 Elsevier Science (USA)

Key Words: **hepatotoxin; nodularin; flounder; mussel; clam; Baltic Sea; bioaccumulation; biotransformation.**

INTRODUCTION

The filamentous cyanobacterium *Nodularia spumigena* occurs regularly in the Baltic Sea during the summer (Sivonen *et al.*, 1989, Kononen *et al.*, 1993, Kankaanpää *et al.*, 2001b). Nodularin (NODLN), produced by the pelagic *N. spumigena*, is a cyclic pentapeptide hepatotoxin (Fig. 1) (Rinehart *et al.*, 1988). Here NODLN refers to cyclo(-D-*erythro*-β-methylAsp(iso-linkage)-L-Arg-Adda-D-Glu(iso-linkage-*N*-methyldehydrobutyrine). Most of the *N. spumigena* blooms in the Baltic Sea have been proven hepatotoxic by mouse bioassay (Sivonen *et al.*, 1989). NODLN concentrations are usually less than 2.4 mg/g dry wt *N. spumigena*, although higher levels have been observed (Sivonen *et al.*, 1989, Kononen *et al.*, 1993). In fact, recent genetic research has indicated that all pelagic *N. spumigena* populations are toxic (Laamanen *et al.*, 2001). In summer 2000, cyanobacterial bloom in the northern Baltic Sea was dominated by *Aphanizomenon flos-aquae* with healthy and decaying *N. spumigena*. Phytoplankton was always hepatotoxic (0.15–8.7 g hepatotoxin/kg phytoplankton dry wt) as assayed with ELISA and HPLC (Kankaanpää *et al.*, 2001c).

In the liver, NODLN and cyclic heptapeptide microcystins (MCs) inhibit the key regulatory enzymes:

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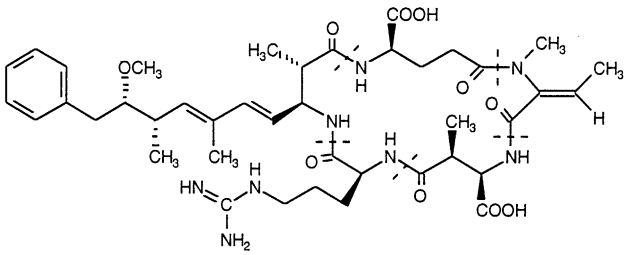

FIG. 1. Chemical structure of nodularin (Rinehart *et al.*, 1988).

serine–threonine protein phosphatases (PPs) 1, 2A, and 3 (Yoshizawa *et al.*, 1990, Honkanen *et al.*, 1994, Annila *et al.*, 1996). These phosphatases participate, e.g., in carbohydrate and lipid metabolism and regulation of apoptosis and cell divisions rates (Annila *et al.*, 1996). NODLN has also been characterized as a carcinogen (Ohta *et al.*, 1994) and apoptogen (Fladmark *et al.*, 1998). NODLN may affect Baltic marine biota; e.g., salmon hepatocytes are reported more sensitive to NODLN than rat hepatocytes (Fladmark *et al.*, 1998). The acute LD_{50} of NODLN is about 50 μg/kg body wt (mouse, ip) (Eriksson *et al.*, 1988; Rinehart *et al.*, 1994).

Toxic *N. spumigena* blooms have caused animal deaths in different parts of the Baltic Sea (Edler *et al.*, 1985; Nehring, 1993; Kononen and Elbrächter, 1996). Recent studies have indicated that cyanobacterial hepatotoxins accumulate in blue mussels and some Baltic fish species, e.g., flounder (*Platichthys flesus*), Atlantic cod (*Gadus morhua*), and blue mussel (*Mytilus edulis*) (Sipiä *et al.*, 2001a, b), and threespine stickleback (*Gasterosteus aculeatus*) (Kankaanpää *et al.*, 2001b).

Exposure to NODLN may contribute to the observed occurrence of liver tumors in Baltic flounder (Wiklund and Bylund, 1994). Thus, accumulation of cyanobacterial hepatotoxins in mussels and flounder (as mussel consumers) is of interest.

In this study, the bioaccumulation and detoxication of cyanobacterial hepatotoxins (mainly NODLN) in flounder, mussels, and clams from the northern Baltic Sea using ELISA, LC coupled with a Finnigan MAT Classic LCQ ion-trap mass spectrometry (LC-MSn), and MALDI-TOF-MS techniques.

MATERIALS AND METHODS

Reagents. Organic solvents (methanol, *n*-butanol) were of analytical or chromatographic grade and purchased from Fisher Scientific (Leicestershire, UK) and Merck (Darmstadt, Germany). Water was purified with Milli-Q_{PLUS} apparatus (Millipore, Bedford, MA). NODLN was purchased from Calbiochem (La Jolla, CA).

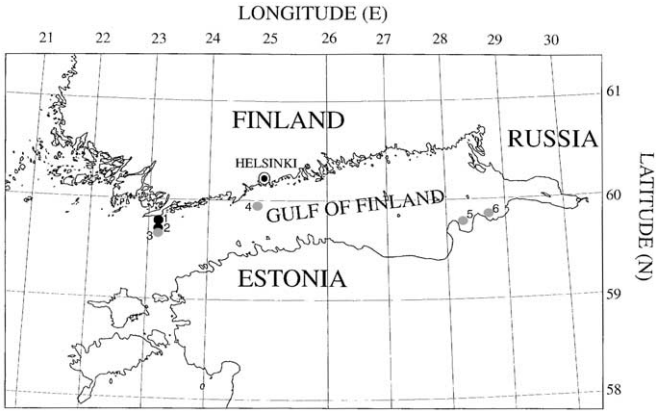

FIG. 2. Sampling locations in the Gulf of Finland in August 1999 and June–September 2000. Site 1 = Sundholm Bay (mussel and flounder samples). Site 2 = S/S Eira shipwreck (mussel samples). Sites 3–6 = mussel samples.

Flounder sampling. Flounder (*Platichthys flesus*) were collected from depths of 10–20 m on 21 August 1999, 25 July, and 25 August 2000 in the western Gulf of Finland, northern Baltic Sea (59° 50. 42′ N, 23° 16. 05′ E, Fig. 2, site 1). Each catch consisted of 30 flounder. The average age of flounder was 5–7 years. Livers and muscles of the flounder were isolated on landing, pooled, and stored at −30°C. Reference flounder, used as a negative control for ELISA, were collected from the Norwegian Sea (Indre Kårvik, Ringvassöja, Tromsö, Norway) during May 1999. No hepatotoxic cyanobacterial blooms have been reported in this area.

Mussel sampling. Blue mussel (*Mytilus edulis*) samples (200 specimens in each sample set) were collected in the western Gulf of Finland by scuba diving from: (1) a steel cage in the Sundholm island at a depth of 3 m on 20 August 1999 (59° 50. 42′ N, 23° 16. 05′ E, Fig. 2, site 1); (2) an enclosure (a rocky bottom) at depths of 3–5 m (site 1); and (3) an artificial reef (wreck s/s Eira, 59° 45 28′ N, 23° 16 11′ E, Fig. 2, site 2) at depths of 25–30 m between June and September 2000.

A total of 200 blue mussels were also collected from two locations between August and October 2000: (1) south of the town Hanko at depths of 5 and 20 m (59° 42′ N, 23° 10.82′ E, Fig. 2, site 3) in the western Gulf of Finland, and (2) south of the city of Helsinki at a depth of 7 m (59° 56.9′ N, 24° 55.8′, Fig. 2, site 4) in the central Gulf of Finland. Furthermore, 50–100 clams and mussels (*M. balthica*, *D. polymorpha*) were collected in each sample set (total five) at a depth of 12 m from Russian waters (Fig. 2, sites 5 and 6) in the eastern Gulf of Finland on 1–4 August 2000. Soft tissue in each sample set was isolated and pooled. Samples were stored at −20 to 25°C. Reference mussels used as negative controls for ELISA were collected from the Atlantic (Vigo Bay,

western Spain), where no NODLN- or MC-producing cyanobacteria have been reported.

Pretreatment of samples. Pooled flounder liver and muscle tissues + pooled mussel and clam soft tissues were homogenized, frozen at $-30°C$, and freeze-dried (Edwards, 12 K Super Modulyo Freeze Drier, West Sussex, England). The dry tissues were homogenized further using a planetary mill (Fritz, Planetary Mill Pulverisette 5, Idar-Oberstein, Germany) and stored at $-30°C$ until extraction.

Extraction and concentration of samples for toxin analysis with ELISA. Dry 1-g samples were extracted with 30 ml of 100% MeOH (Carmichael, 1995; Williams *et al.*, 1997a) in a Branson 3200 (Dansbury, CT) ultrasonic bath (50–60°C) for 8 h with occasional shaking. The extracted sample was centrifuged at 1300*g* for 15 min in a Megafuge 2.0 R (Heraeus Sepatech, Osterode, Germany). The supernatant was collected and stored overnight in a refrigerator. A supernatant was concentrated *in vacuo* (50°C) and under a nitrogen stream to 350 μl and centrifuged at 5200*g* for 15 min. One hundred microliters of the concentrated sample extract was diluted with 900 μl (1 : 10) of Milli-Q water. The final sample was clarified using Millipore (Bedford, MA) Millex-HV filters (pore size 0.45 μm, diameter 4 mm). Sample solutions were analyzed immediately after filtration.

The mussel sample (20 August 1999) and one flounder sample (21 August, 1999) were analyzed using LC-MS/MS. MALDI-TOF-MS was applied to a mussel tissue sample from 23 August 2000. For these analyses the samples were extracted with water : methanol : *n*-butanol (75 : 20 : 5). The concentrated extract (350 μl) was cleaned up using Bakerbond C_{18} solid-phase extraction cartridges (500 mg; Baker, Phillipsburg, NJ), which had been activated with 10 ml of 100% methanol followed by 10 ml of Milli-Q water. Then extracts were adsorbed into the cartridges and cartridges were flushed with 20 ml of water and 15 ml of 10% methanol. The samples were eluted with 20 ml of 100% methanol. Finally, the toxin-containing fraction was eluted with 20 ml of 100% methanol and evaporated to dryness.

LC-UV-MS/MS studies on MCs and NODLN were carried out on a HP 1100 series HPLC system with PDA-UV detector (G1315A, Hewlett–Packard) linked with a Finnigan MAT LCQ tandem mass spectrometer (ThermoFinnigan, San Jose, CA) equipped with an electrospray ion-spray (ESI) interface. The mass spectrometer was tuned using MC-LR, with a flow injection at 3 μl/min and an optimized capillary temperature of 220°C with a voltage of 3.0 V. The MCs are separated on a C_{18} Luna(2) column (5 μm, 150 × 2.0 mm, Phenomenex, Macclesfield, UK) at 40°C (5 μl injection). A gradient elution of acetonitrile/water (30/70 with 0.05% TFA) to 100% acetonitrile (with 0.05% TFA) over 42 min is used at a flow rate of 0.2 ml/min. MCs and NODLN were determined by their retention times and/or characteristic mass spectra.

The MALDI-TOF-MS instrument, equipped with a nitrogen laser operating at 337 nm and a two-state ion source operating in the delayed extraction mode (PerSeptive Biosystems, Framingham, MA, is used for characterization of the cyanobacterial hepatotoxin and detoxification conjugates. The samples are subjected to MALDI-TOF using the dried droplet method (Karas and Hillenkamp, 1988). All measurements are performed at an acceleration voltage of 20 kV (delay 200 ns) using the reflector mode, allowing determination of monoisotopic mass values. Each spectrum or measurement refers to the sum of 100–200 individual laser shots (Pflugmacher *et al.*, 1998). Time-of-fight (to MS detector) of splitting ions correlates directly with the atomic mass so that ions of heavier atomic mass have longer TOF values. Finally, MC/NODLN and their conjugates are identified with MS detector.

ELISA was performed using the EnviroLogix plate kit (EnviroLogix Inc.; Portland, ME) according to the manufacturer's instructions. The plates were read at 450 nm with a micro-plate reader (Bio-Rad, Benchmark Microplate Reader, CA). The plate kit used shows good cross-reactivity with MCs and NODLN and a detection limit of 0.2 ng/ml for NODLN. All tissue sample extracts were analyzed in triplicate and NODLN solutions of 0.25 to 1.5 ng/ml (in 10% MeOH) were used for calibration. The results are expressed in NODLN equivalents. Additionally, negative controls were run using the EnviroLogix protocol.

RESULTS AND DISCUSSION

ELISA indicated that mussels, clams, and flounder contained cyanobacterial hepatotoxins in August 1999 and June–October 2000. The reference flounder or mussel samples did not contain significant amounts of NODLN or MCs (< 10 μg/kg dry wt). Thus toxin content in mussels collected in June also exceeded this reference level. This suggests that portions of the toxins are stored in the mussels during the winter. In the case of flounder liver the highest hepatotoxin concentration of 410 ± 12 μg/kg dry wt (Fig. 3) occurred in samples collected from Sundholm Bay (site 1) on 25 August 2000. Liver-bound NODLN content in flounder caught during August was slightly higher in comparison to toxin content of samples collected on 25 July (Fig. 3). Hepatotoxin concentrations in those flounder samples were of the same order of magnitude as those of livers of flounder caught during July (137 ± 7 μg/kg dry wt) and August (399 ± 5 μg/kg dry wt) in 1999 from the Sundholm Bay area (Sipiä *et al.*, 2001b). However, the maximum hepatotoxin concentration was clearly higher than in flounder studied between 1996 and 1998 (30–140 μg/kg dry wt) (Sipiä *et al.*, 2001a). These differences may be caused by temporal differences in sampling as most of the flounder in the earlier study were caught in September. Hence, some of the toxin may have degraded or detoxicated from the tissues prior to

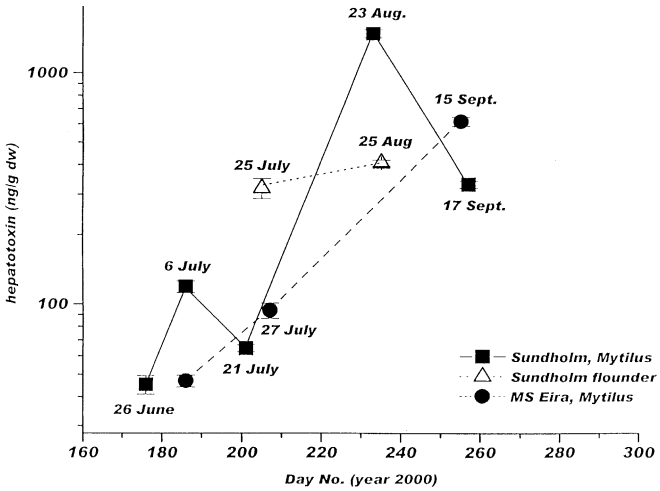

FIG. 3. Temporal accumulation of total cyanobacterial hepatotoxins (mainly nodularin) in blue mussels and flounder from the western Gulf of Finland (sites 1 and 2) in June–September 2000.

sampling. Also, a different extraction solvent (water : methanol : *n*-butanol 75 : 20 : 5 v/v/v) was used earlier (Sipiä *et al.*, 2001a). No hepatotoxins have been detected in flounder muscle. This is in accordance with the observations of Bury *et al.* (1998) who reported that the majority of absorbed MC-LR accumulated in rainbow trout liver.

In the case of mussels, the highest hepatotoxin concentration of 1490 ± 60 μg/kg dry wt was found in mussels collected from Sundholm Bay (Fig. 3) on 23 August 2000. The presence of NODLN in this sample was also confirmed by MALDI-TOF-MS measurements (*m*/*z* 824.904) (Fig. 4). In addition, MALDI-TOF-MS revealed that GSH conjugates were present in the same mussel sample (*m*/*z* 1131.56) (Fig. 4). This finding is evidence of detoxication of NODLN in Baltic biota as found earlier in terrestrial and freshwater environments (Pflugmacher *et al.* 1998). Hepatotoxin content of mussel samples increased about 30-fold from June to August and 20-fold from July to August. Later, hepatotoxin content decreased about 4.5-fold from August to September, which was evidence of at least partial depuration or detoxication in samples. Temporal accumulation of cyanobacterial hepatotoxin in mussel samples was analogous to that of mussels collected from Sundholm Bay in June–September 1999 (Sipiä *et al.*, 2001b). ELISA measurements showed that the mussels collected in the western Gulf of Finland on 20 August 1999 contained greater than 2 mg/kg dry wt NODLN, while livers of flounder caught in the western Gulf of Finland on 21 August 1999 contained 399 ± 5 μg/kg dry wt NODLN (ELISA). Also, LC-MS/MS measurements confirmed that NODLN was the only hepatotoxin in mussel and flounder samples collected from the Gulf of Finland in August 1999 (Fig. 5) (Kankaanpää *et al.*, 2001c).

MCs, closely related to NODLN with respect to structure and function, have been shown to be metabolized in zebra fish (*Danio rerio*) (Wiegand *et al.*, 1999), rodents (Kondo and Harada, 1996) and plants (Pflugmacher *et al.*, 1999) by detoxication enzymes, namely, soluble glutathione

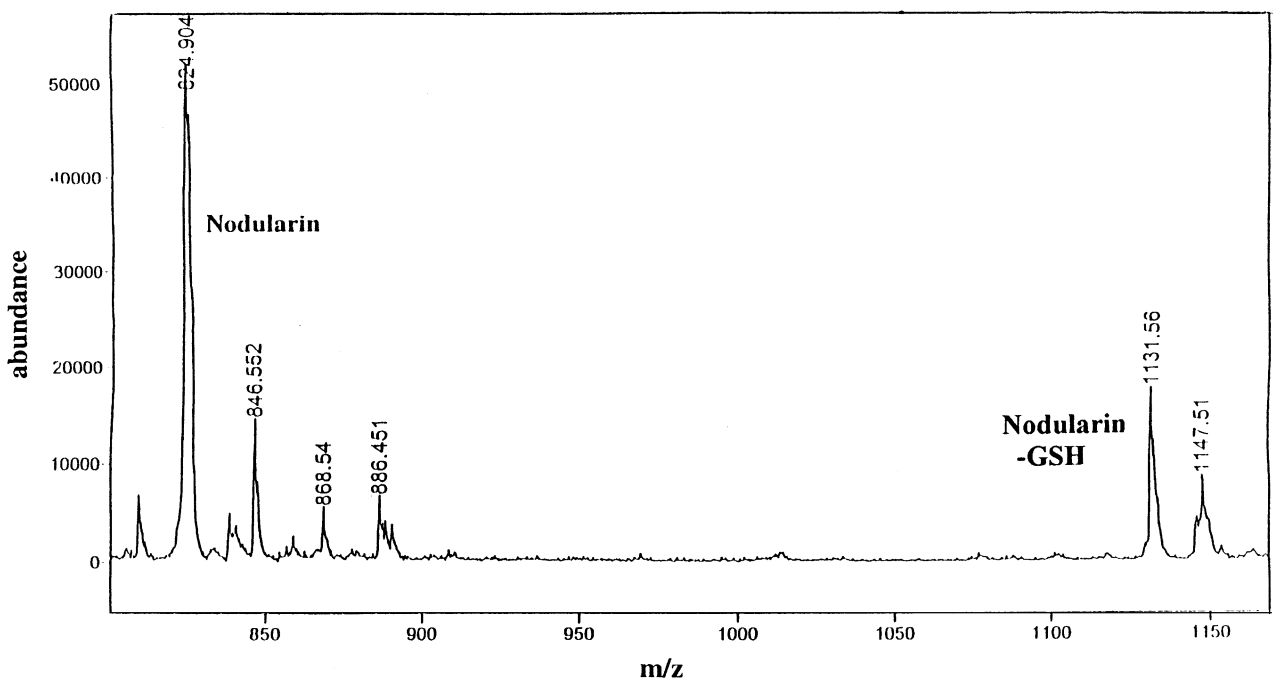

FIG. 4. MALDI-TOF-MS measurement of the soft tissue of mussels collected on 23 August 2000.

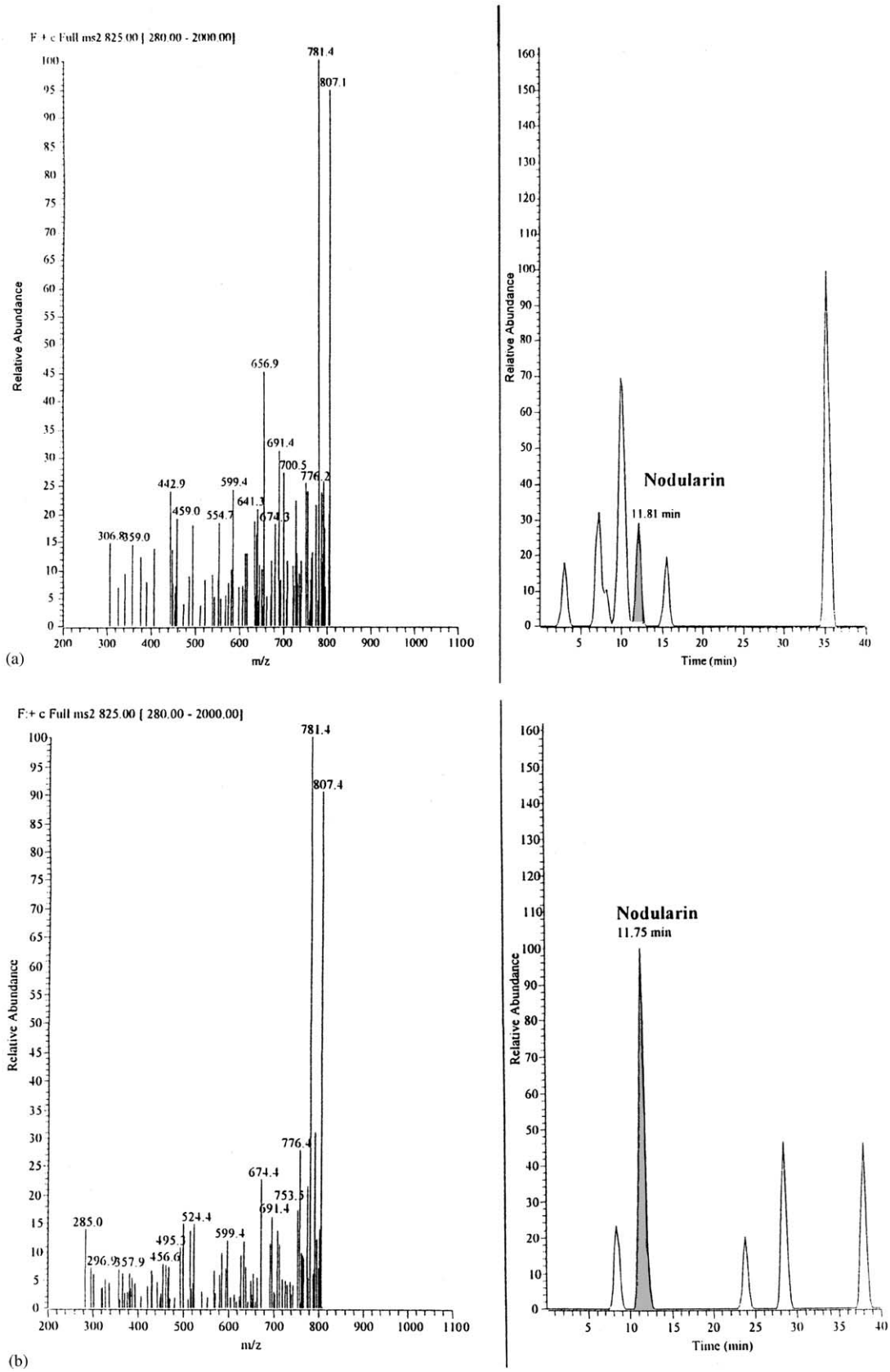

FIG. 5. LC-MS/MS measurements of mussel soft tissue collected on 20 August 1999 (a) and liver of flounder caught on 21 August 1999 (b). Both samples were collected in the western Gulf of Finland.

S-transferases (sGSTs) (Habig *et al.*, 1974; Pflugmacher *et al.*, 1998). sGSTs catalyze the addition of MCs to the SH (thiol) group of the tripeptide glutathione (GSH). This can neutralize the electrophilic sites of MCs and increase water solubility and thus enhance excretion of degradation products, like GSH conjugates of MC-LR (Pflugmacher *et al.*, 1998; Wiegand *et al.*, 1999). Conjugation of GSH and cysteine to MC-LR has been observed (Wiegand *et al.*, 1999).

Falconer *et al.* (1992) reported that hepatotoxicity in Australian blue mussels increased during a dense bloom of *N. spumigena* and decreased rapidly after the bloom. The findings of this study and that of 1999 (Sipiä *et al.*, 2001b) are analogous to those of Falconer *et al.*, with only small amounts of *N. spumigena* present in the water. Depuration of the Baltic cyanobacterial hepatotoxins is rather rapid, but not complete. Williams *et al.* (1997b) also reported that mussels exposed to MCs rapidly cleared covalently bound MCs when transferred to toxin-free seawater and this may be a result of ongoing detoxication activity in the mussels.

In contrast to mussels collected from the surface layer (at depths of 3–5 m) in Sundholm Bay (Fig. 2, site 1), the hepatotoxin levels in mussels collected from the wreck (Fig. 2, site 2) at 25–30 m continued to increase between 27 July and 15 September 2000 (Fig. 3). On 15 September the wreck mussels contained greater than 10-fold more hepatotoxin (about 620 ± 30 μg/kg dry wt) than mussels from the same site on 6 July (Fig. 3). This indicates that cyanobacterial hepatotoxins continuously reach the sea floor water layers. Similar time-dependent accumulation was also observed in mussels collected from the same wreck (Fig. 2, site 2) in July–September 1999 (Sipiä *et al.*, 2001b). Whether this lagged bioaccumulation occurs via particle-bound NODLN or NODLN in water phase is unclear. Thus, the cyanobacterial hepatotoxins accumulated rapidly in the shallow-lying mussels, but it took more time for the toxins to reach the mussels at 25–30 m.

Heiskanen (1998) reported that only a very small fraction of cyanobacterial blooms reach sediments. Other studies have also indicated a modest sedimentation rate of the summer cyanobacterial production in the Gulf of Finland (Sellner, 1997). However, filaments of *N. spumigena* have recently been observed in sediment in the Gulf of Finland (Kankaanpää *et al.*, 2001b). Additionally, sedimentation of *N. spumigena* has been traced using sediment traps at a depth of 25 m in the western Gulf of Finland (Kankaanpää *et al.*, 2001a).

Cyanobacterial hepatotoxins were also found in mussels collected from another site in the western Gulf of Finland (Fig. 2, site 3) on 11 August 2000 at depths of 5 and 20 m. Results revealed that the mussels from the surface layer (5 m) contained greater than 3-fold more hepatotoxins (about 135 μg/kg dry wt) than the mussels from 20 m. In addition, the mussels collected from the central Gulf of Finland (7-m depth, Fig. 2, site 4) on 14 October 2000 contained ca. 130 μg hepatotoxins/kg dry wt. Furthermore, clams and mussels (*M. balthica*, *D. polymorpha*) collected at 12 m in the eastern Gulf of Finland (Fig. 2, sites 5 and 6) on 1–3 August 2000 contained cyanobacterial hepatotoxins. The hepatotoxin levels in clams exceeded the Atlantic reference levels ranging from about 100 ± 10 to 130 ± 10 μg/kg dry wt.

CONCLUSION

The results of this study indicate that cyanobacterial hepatotoxins constantly bioaccumulate in benthic organisms in the Gulf of Finland. This accumulation in mussels and further in flounder takes place even in the presence of low cell numbers of toxic cyanobacteria. Thus, long-term effects of hepatotoxic NODLN to Baltic biota must be taken into account since hepatotoxins may have chronic toxic effects such as tumor promotion.

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REFERENCES

Annila, A., Lehtimäki, J., Mattila, K., Eriksson, J. E., Sivonen, K., Rantala, T. T., and Drakenberg, T. (1996). *J. Biol. Chem.* **271,** 16695–16702.

Bury, N. R., Newlands, A. D., Eddy, F. B., and Codd, G. A. (1998). *Aquat. Toxicol.* **42,** 139–148.

Carmichael, W. W. (1995). In *Manual on Harmful Marine Microalgae* (G. M. Hallegraeff, D. M. Andersson, and A. D. Cembella, Eds.), pp. 165–177.

Edler, L., Fernö, S., Lind, M. G., Lundberg, R., and Nilsson, P. O. (1985). *Ophelia* **24,** 103–109.

Eriksson, J. E., Meriluoto, J. A. O., Kujari, H. P., Österlund, K., Fagerlund, K., and Hällblom, L. (1988). *Toxicon* **26,** 161–166.

Falconer, I. R., Choice, A., and Hosja, W. (1992). *Environ. Toxicol. Water Qual.* **7,** 119–123.

Fladmark, K. E., Serres, M., Larsen, N., Yasumoto, T., Aune, T., and Doskeland, S. (1998). *Toxicon* **36,** 1101–1114.

Habig, W. H., Pabst, M. J., and Jakoby, W. B. (1974). Glutathione *S*-transferases. *J. Biol. Chem.* **249,** 71530–71539.

Heiskanen, A. S. (1998). *Monogr. Bor. Environ. Res.* **8,** 61.

Honkanen, R. E., Codispoti, B. A., Tse, K., and Boynton, A. L. (1994). *Toxicon* **32,** 339–350.

Kankaanpää, H., Sipiä, V., and Huttunen, M. (2001a). In *The 5th International Conference on Toxic Cyanobacteria, Noosa, Queensland, Australia, July 15–20, 2001.*

Kankaanpää, H., Sipiä, V. O., Kuparinen, J. S., Chizmar, J. L., Carmichael, and Wayne, W. (2001b). *Phycologia* **40,** 268–274.

Kankaanpää, H., Sipiä, V., Pflugmacher, S., Furey, A., James, K., Lehane, M., Huttunen, M., and Kuparinen, J. (2001c). In *The 5th International Conference on Toxic Cyanobacteria, Noosa, Queensland, Australia, July 15–20, 2001.*

Karas, M., and Hillenkamp, F. (1988). *Anal. Chem.* **60,** 2299–2304.

Kondo, F., and Harada, K.-I. (1996). *J. Mass Spectr. Soc. Jap.* **44,** 355–376.

Kononen, K., and Elbrächter, M. (1996). In *Warnsignale aus der Ostsee: Wissentschaftliche Fakten* (Lozan *et al.*, Eds.), pp. 138–141. Parey, Berlin.

Kononen, K., Sivonen, K., and Lehtimäki, J. (1993). In *Toxic Phytoplankton Blooms in the Sea* (T. J. Smayda and Y. Shimizu, Y. Eds.), pp. 269–273.

Laamanen, M., Gugger, M., Lehtimäki, J., Haukka, K., and Sivonen, K. (2001). Diversity of toxic and non-toxic *Nodularia* isolates (cyanobacteria) and filaments from the Baltic Sea. *Appl. Environ. Microbiol.* **67,** in press.

Nehring, S. (1993). *J. Plankton Res.* **15,** 867–872.

Ohta, T., Sueoka, E., Iida, N., Komori, A., Suganuma, M., Nishiwaki, R., Tatematsu, M., Kim, S-J., Carmichael, W. W., and Fujiki, H. (1994). *Cancer Res.* **54,** 6402–6406.

Pflugmacher, S., Geissler, K., and Steinberg, C. (1999). *Ecotoxicol. Environ. Saf.* **42,** 62–66.

Pflugmacher, S., Wiegand, C., Oberemm, A., Beattie, K. A., Krause, E., Codd, G. A., and Steinberg, C. E. W. (1998). *Biochim. Biophys. Acta* **1425,** 527–533.

Rinehart, K. L., Harada, K., Namikoshi, M., Chen, C., and Harris, C. E. J. (1988). *J. Am. Chem. Soc.* **110,** 8557–8558.

Rinehart, K. L., Namikoshi, M., and Choi, B. W. (1994). *J. Appl. Phys.* **6,** 159–176.

Sellner, K. (1997). *Limnol. Ocean.* **42,** 1089–1104.

Sipiä, V. O., Kankaanpää, H. T., Lahti, K., Carmichael, W. W., and Meriluoto, J. A. O. (2001a). *Environ. Toxicol.* **16,** 121–126.

Sipiä, V. O., Kankaanpää, H. T., Lahti, K., Flinkman, J., and Meriluoto, J. A. O. (2001b). *Environ. Toxicol.* **16,** 330–336.

Sivonen, K., Kononen, K., Carmichael, W. W., Dahlem, A. M., Rinehart, K. L., Kiviranta, J., and Niemelä, S. I. (1989). *Appl. Environ. Microbiol.* **55,** 1990–1995.

Wiegand, C., Pflugmacher, S., Oberemm, A., Meems, N., Beattie, K. A., Steinberg, C. E. W., and Codd, G. A. (1999). *Environ. Toxicol.* **14,** 89–95.

Wiklund, T., and Bylund, G. (1994). In *Diseases and Parasites of Flounder* (*Platichthys flesus*) *in the Baltic Sea* (Bylund *et al.*, Eds.), BMB Publ. 15, pp. 49–52.

Williams, D. E., Dawe, S., Kent, M., Andersen, R., Graig, M., and Holmes, C. (1997a). *Toxicon* **35,** 1617–1627.

Williams, D. E., Graig, M., Dawe, S., Kent, M., Holmes, C., and Andersen, R. (1997b). *Chem. Res. Toxicol.* **10,** 463–469.

Yoshizawa, S., Matsushima, R., Watanabe, M., Harada, K., Ichihara, A., Carmichael, W. W., and Fujiki, H. J. (1990). *Cancer Res. Clin. Oncol.* **116,** 609–614.